Information_Security

Unit1:

Security Concepts: Introduction, The need for security, Security approaches, Principles of security, Types of Security attacks, Security services, Security Mechanisms, A model for Network Security Cryptography Concepts and Techniques: Introduction, plain text and cipher text, substitution techniques, transposition techniques, encryption and decryption, symmetric and asymmetric key cryptography, steganography, key range and key size, possible types of attacks.

Unit2:

Symmetric key Ciphers: Block Cipher principles, DES, AES, Blowfish, RC5, IDEA, Block cipher operation, Stream ciphers, RC4. Asymmetric key Ciphers: Principles of public key cryptosystems, RSA algorithm, Elgamal Cryptography, Diffie-Hellman Key Exchange, Knapsack Algorithm.

Unit 1 whole, in unit 2 study Symmetric key Ciphers: Block Cipher principles, DES, AES, Comparison of Blowfish, RC5, IDEA, Block cipher operation, Diffie Hellman key Exchange Algorithm

Euclidean Algorithm for GCD, Extended Euclidean for inverse modulus..Chinese remainder, Exponents over modulus..